

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

2. Requirement Analysis Phase

The Requirement Analysis Phase focuses on identifying what the FitBuddy application should do and what resources are required to develop it. The requirements were divided into functional requirements and non-functional requirements.

The primary functional requirement is that FitBuddy should collect user fitness information through a web interface. The user provides a unique User ID, name, age, weight, fitness goal, and workout intensity. The application validates this information before generating the plan.

The next requirement is AI-based workout generation. FitBuddy uses a Gemini model to generate a structured 7-day workout plan. The plan should contain appropriate exercises along with relevant workout details such as sets, repetitions or duration, warm-up, rest periods, and cool-down.

Another functional requirement is nutrition and recovery guidance. Along with the workout plan, FitBuddy should generate a concise nutrition and recovery recommendation according to the user's fitness goal.

The system also requires a feedback mechanism. After viewing the generated workout plan, users can submit feedback. The application retrieves the user's original plan and user information and sends the relevant information to the AI system. A new updated workout plan is then generated based on the feedback.

The application also requires database storage. User details, original workout plans, updated plans, and nutrition recommendations are stored so that the information can be retrieved later.

The major requirements identified are:

- Input Requirements: User ID; Name; Age; Weight; Fitness goal; Workout intensity; User feedback.
- Processing Requirements: Validate user input; Generate workout plan using Gemini; Generate nutrition/recovery recommendation; Process user feedback; Generate updated workout plan; Store information in the database.
- Output Requirements: Personalized 7-day workout plan; Nutrition and recovery recommendation; Updated workout plan after feedback; User information and stored plans.

The main technologies selected for implementation are Python, FastAPI, Jinja2, SQLAlchemy, SQLite, HTML, CSS, and Google's Gemini API. The project also uses environment variables for securely storing configuration information such as the Gemini API key.

The requirement analysis therefore provides a clear specification for developing the FitBuddy application.